



# Controlling properties of metal–polymer soft magnetic composites through microstructural deformation for power inductor applications

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## ABSTRACT

We studied the effect of microstructural deformation on soft magnetic metal powders in a power inductor operating above 1 MHz. In this study, an inductor core was fabricated using Fe-6.5Si powders that exhibit a high electrical resistance and a large magnetic saturation value. This core was formed with a high-density microstructure using nano Fe powders synthesized via pulsed wire evaporation. The permeability was maximized at the content ratio for which the maximum density and packing fraction were observed according to the relationship between permeability and packing fraction outlined in Ollendorff's equation. Experimentally, the highest packing fraction and permeability were observed in the core containing 20 wt% nano Fe powder. Further, the sample with the highest packing fraction showed a relatively low core loss, compared to the other samples. Thus, the inductor core is affected by the intrinsic properties and microstructure of the soft magnetic material used.

## 1 Introduction

The size of the inductor has been miniaturized along with the trend for miniaturizing electronic products. To reduce the size of an inductor, soft magnetic materials that exhibit excellent DC biasing characteristics and a stable magnetic permeability in the high frequency band, and can achieve a high current of 1 MHz or more are required [1–4]. In general, metal powders with high magnetic saturation properties exhibit excellent DC biasing characteristics, because they can maintain inductance at high

currents and have high current capacities [5, 6]. However, compared to ferrites, these metal powders have relatively low electrical resistivities [7, 8]. Therefore, the eddy-current loss increases rapidly in the high-frequency region, limiting the applications of metal powders as electrical devices including power inductors [9]. A high electrical resistivity is essential for reducing the eddy-current loss of materials [10]. To overcome these limitations, Fe-based amorphous powders have been proposed and used as viable candidates for power applications by several researchers because of their high electrical

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resistivity [11–13]. However, amorphous powders have the disadvantage of a low saturation flux density, compared to silicon steel [14]. Further, amorphous materials are difficult to fabricate and are expensive. In contrast, crystalline Fe-based alloys are easy to fabricate and supply. Thus, in our study, we used crystalline Fe–Si powder as the main material because of its higher electrical resistivity compared to that of pure Fe powders.

To improve the magnetic permeability and reduce the total core loss, we used soft magnetic composites (SMCs) [15]. Recently, several studies on various kinds of SMCs, such as FeSiCr [16–18], FeSiAl [19–21], FeNi [22, 23], and amorphous Fe [24–26], have been reported. Using SMC cores with a high packing fraction that takes advantage of different types and sizes of metal powders can also lead to an increase in the permeability and reduce the core loss [27–31]. The core properties such as packing fraction and permeability well satisfy the relationships described by Ollendorff's equation.

In this study, the microstructures of some inductor cores were modified using two different metal powders with two different particle sizes. The Fe–Si-based alloy is also called silicon steel and is widely used as a magnetic material for electric-magnetic and electric-mechanical energy conversion in generators, transformers, and motors [32]. In particular, silicon reduces the core loss by increasing the electrical resistivity [33]. Among the known Fe–Si-based alloys, Fe-6.5Si, which contains 6.5 wt% of Si, was used in this study. According to a previously reported study, the resistivity increases from 48 to 82  $\mu\Omega\cdot\text{cm}$  as the Si content is increased from 3 to 6.5 wt% [34]. Furthermore, the eddy-current losses in the core can be reduced by using soft magnetic materials with high resistivities. Fe-6.5 wt% Si alloy powder, containing spherical particles, with a high resistivity, low crystal anisotropy, and low magnetostriction constant, was used. This Fe-6.5 wt% Si alloy powder was produced by using the gas atomization method exhibited outstanding soft magnetic properties compared to the other Fe–Si-based powder. The permeability of the core shows different properties depending on the size of the Fe-6.5 wt% Si powders used. Thus, Si alloys need to be investigated in further detail for developing high-power devices with high permeabilities, low eddy-current losses, and excellent mechanical properties [35, 36]. A nano Fe powder fabricated using the pulsed wire evaporation (PWE) technique

was used to increase the density and packing fraction of the core. The PWE method employs a conductive metal wire with a uniform diameter; this wire is vaporized using a high voltage over a very short time in a chamber [37, 38] and subsequently precipitated in a water bath placed inside the chamber. The nano-sized Fe powder, having a large surface energy, may be rapidly oxidized through natural firing in air [39, 40]. However, the nano-sized Fe powder prepared by the PWE method may precipitate in the chamber to suppress the natural firing. The dried nano Fe powder forms an oxide layer on the surface, thereby increasing the resistivity [39, 41]. We observed the microstructure of the core, consisting of the nano Fe and Fe-6.5Si powders, and measured the AC and DC magnetic properties and core loss.

Ollendorff's equation shows the relationship between the packing fraction and permeability of the core. We calculated the packing fraction according to the microstructural deformation and measured the permeability of the core, and obtained a value same as the theoretical value derived from Ollendorff's equation. To date, Fe–Si alloys, i.e., silicon steel, have been primarily used as suitable materials in power inductors. However, adding PWE-made nanopowder to improve the core packing fraction has not yet been considered to the best of our knowledge.

## 2 Experimental details

A toroidal core was fabricated to observe the effect of powder size on the permeability and energy loss. The core was made of Fe-6.5Si powders, which have excellent intrinsic soft magnetic properties. Furthermore, we used nano Fe powders to increase the density and packing fraction. The Fe-6.5Si and nano Fe powders were synthesized using the gas atomization and PWE methods, respectively. The average particle sizes of the powders were analyzed using a particle size analyzer (PSA, MKT LA-950), and the morphologies of the powder and oxide layer were observed through a scanning electron microscope (SEM, Hitachi S3500) and transmission electron microscope (TEM, JEM-2100 F). The compositions of the powders were analyzed using energy-dispersive spectroscopy (EDS), and the crystal structures of the powders were analyzed through X-ray diffraction (XRD, Rigaku Miniflex). The Fe-6.5Si and nano Fe powders were mixed with 3.0 wt% of an insulating

epoxy binder, and a toroidal core was prepared using a mold. The outer and inner diameters of the toroidal core were 20 and 13 mm, respectively. Here, 3 g of the mixed powders was used to manufacture the core, and a uniform pressure of  $1.2 \text{ ton/cm}^2$  was applied. Next, the fabricated cores were cured by heat treatment at  $160^\circ\text{C}$  for 1 h, and a Cu wire with a diameter of 0.25 mm was used for the winding on the cores. We measured the relative permeability of the core from 100 kHz to 110 MHz using an impedance analyzer (Hewlett Packard 4294 A). The sample's core loss was measured using a B–H analyzer (IWATSU SY-8218), and the DC magnetic properties such as magnetic saturation and coercivity were measured using a vibrating sample magnetometer (VSM, Lake Shore 7404).

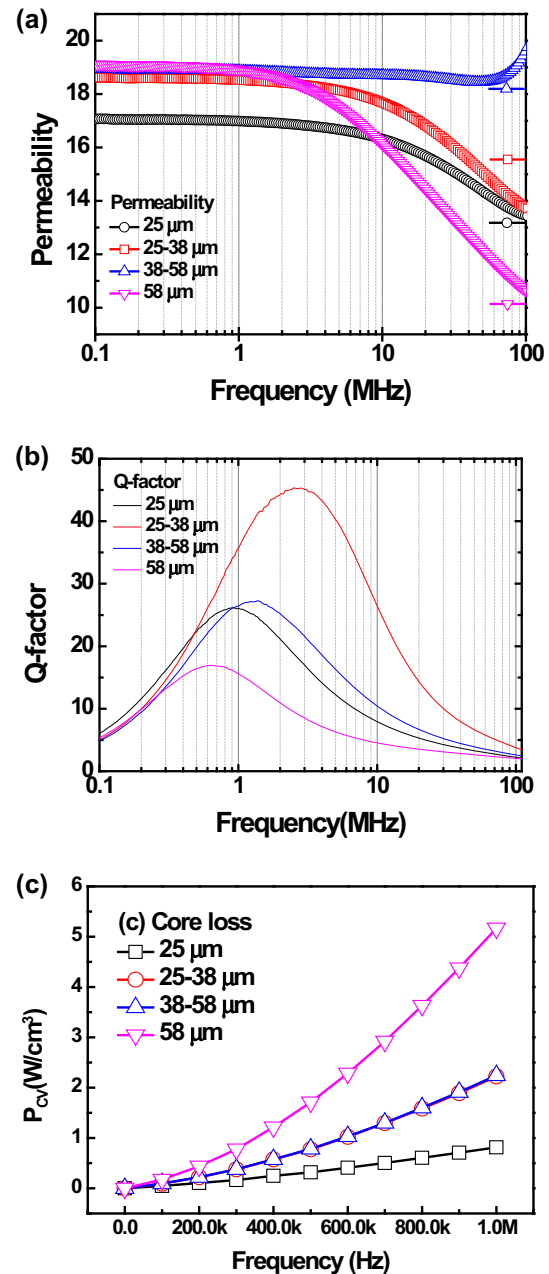
### 3 Results and discussion

#### 3.1 Characteristics of the powders

The Fe-6.5Si alloy powder was used to obtain a high permeability and low core loss for a power inductor. We prepared four sizes of the Fe-6.5Si powders ( $< 25 \mu\text{m}$ ,  $25\text{--}38 \mu\text{m}$ ,  $38\text{--}58 \mu\text{m}$ , and  $\geq 58 \mu\text{m}$ ) to confirm their outstanding soft magnetic properties. Figure 1 shows the effect of powder particle size on permeability, Q-factor, and core loss. At 1 MHz, the permeabilities of the inductor core with particle sizes of  $\leq 25$ ,  $25\text{--}38$ ,  $38\text{--}58$ , and  $\geq 58 \mu\text{m}$  were 16, 18.5, 18.8, and 18.9 H/m, respectively; thus, as the particle size was increased, the permeability gradually increased as well.

The Q-factor indicates the quality of the frequency-selective characteristic at resonance. At 1 MHz, the Q-factors of the inductor core with particle sizes of  $\leq 25$  and  $25\text{--}38 \mu\text{m}$  were 26 and 35.7, respectively, while the frequency corresponding to the maximum Q-factor (45) increased to 2.5 MHz. Further, the Q-factor of the inductor core with a particle size in the range of  $38\text{--}57 \mu\text{m}$  was 26.4 at 1 MHz, and the maximum Q-factor was 27 at 1.4 MHz. However, the value of the Q-factor of the inductor core with a particle size of  $\geq 58 \mu\text{m}$  drastically reduced to 15 at 1 MHz. Further, in the low-frequency band, the maximum Q-factor shifted to a lower frequency, i.e., the Q-factor was 16 at 600 kHz.

Furthermore, the core loss is as important as the permeability and Q-factor of power inductors. Even if

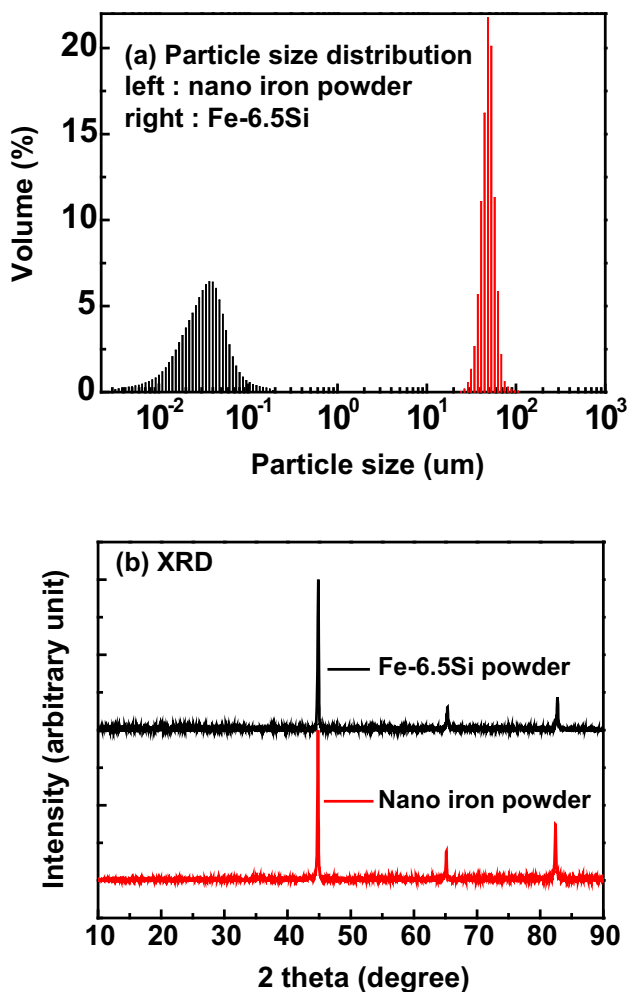


**Fig. 1** a Permeability, b Q-factor, c core loss of Fe-6.5Si metal powders according to the particle size

the permeability and Q-factor are high, a large core loss is detrimental to the applications of an inductor. Comparing the core loss at 1 MHz, the core with a particle size of  $25 \mu\text{m}$  indicates the lowest core loss of approximately  $814 \text{ mW/cm}^3$ , whereas that with particle size of  $58 \mu\text{m}$  exhibits the highest core loss of approximately  $5168 \text{ mW/cm}^3$ . This result confirms that the particle size affects the energy loss.

Figure 2 shows the particle size distribution of the Fe-6.5Si and nano Fe powders used in the inductor. The Fe-6.5Si powders exhibited a size distribution from 20 to 90  $\mu\text{m}$ , and were sieved. The nano Fe powders exhibited a size distribution from 1 to 100 nm, and the particles in the range of 35–50 nm showed the maximum distribution. Moreover, the XRD peak of each powder was similar to the Fe peak of the ICDD card. The Si peak of Fe-6.5Si was not observed because it is very small. [32]

However, the Si component can be observed through EDS mapping as shown in Fig. 3. The nano Fe powder was also observed to contain a small oxygen component, through EDS mapping, and through TEM, it was confirmed that the oxygen layer was an oxide layer of approximately 5 nm thickness. This oxide layer exhibited typical insulating



**Fig. 2** **a** Particle size distribution, **b** XRD peak of Fe-6.5Si alloy powder and nano iron powder

properties in the core-shell structure as well as prevented spontaneous ignition of the nano powder. [42]

### 3.2 Characteristic of the soft magnetic composite core

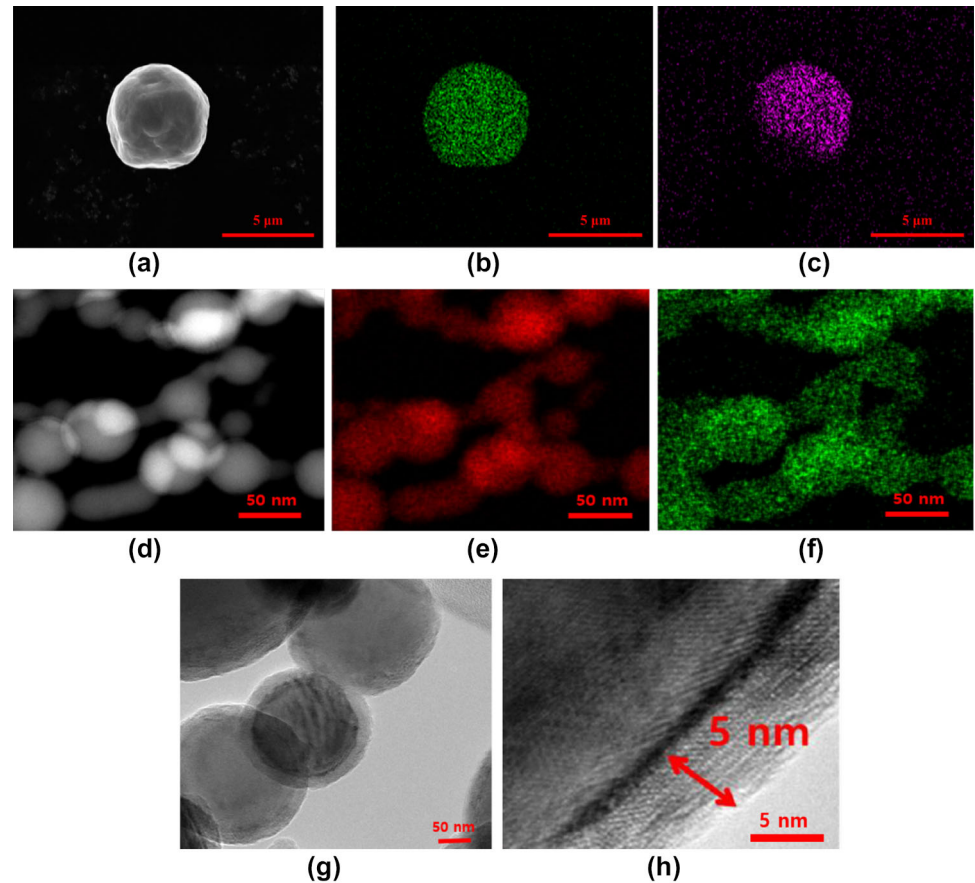
Figure 4a shows the density and packing fraction of the core when the nano Fe powder is added. Evidently, the density and packing fraction are entirely dependent on the amount of the added nano Fe powder. The density of the core without the nano Fe powder is  $5.14 \text{ g/cm}^3$ , whereas that with 20 wt% nano Fe powder is  $6.00 \text{ g/cm}^3$ . Figure 4b shows the microstructure in which the nano Fe powder fills the hole. The Fe-6.5Si and nano Fe powders are uniformly distributed in the cross section of the core containing 20 wt% nano Fe powder.

Figure 5a shows the DC magnetic characteristics of the core, analyzed using a VSM. The hysteresis loop, which indicates to the magnetic properties of the core, depends on the density and packing fraction. In our case, the hysteresis loop, expressed as the magnetization value per unit mass of the core, shows a decrease in the magnetic saturation and an increase in the coercivity when the amount of nano Fe powder is increased. However, the hysteresis loop expressed as the magnetization value per unit volume of the core shows different results, i.e., a large magnetic saturation is observed in the core containing 20 wt% nano Fe powder. This indicates that inductors with a high density and packing fraction in the same volume are more easily magnetized, implying that permeability is an important characteristic in inductors and can improve their energy efficiency in power applications such as DC–DC conversion. [43]

Figure 6a shows the permeability of the core and its frequency dependence from 100 kHz to 110 MHz. The resonant frequencies of the 20 wt% and 40 wt% nano Fe powder inductor cores with high packing fractions shift to less than 100 MHz. The 60 wt%, 80 wt%, and 100 wt% nano Fe powder inductor cores with low packing fractions show a shift in the resonant frequency above 100 MHz. Figure 6b shows the permeability of the core at 1 and 3 MHz. These are the frequency bands usually used in portable electronic devices [27]. The permeability of the core without the nano Fe powder is 18 H/m, and it constantly increases as the content of the nano Fe powder is increased. The 20 wt% nano Fe powder core with a high packing fraction shows the maximum



**Fig. 3** SEM image of **a** Fe-6.5Si, EDS mapping of **b** Fe distribution, **c** Si distribution, SEM image of **d** nano Fe powder, EDS mapping of **e** Fe distribution, **f** Oxygen distribution, **g** TEM of nano Fe powder, and **h** oxygen layer of nano Fe powder



permeability of approximately 35 H/m. However, adding more nano Fe powder produced a negative effect on the permeability of the core. In the 100 wt% nano Fe powder inductor core, the permeability was significantly reduced to 12 H/m. Ollendorff's equation (Eq. 1) shows the correlation between packing fraction and permeability as [44–46]

$$\mu = \mu_0 + \frac{\eta\mu_0(\mu_m + \mu_0)}{N(1 - \eta)(\mu_m - \mu_0) + \mu_0}, \quad (1)$$

where  $\eta$  is the volume packing fraction,  $N$  is the coefficient of the demagnetizing field ( $0 \leq N \leq 1$ ),  $\mu_m$  is the permeability of the material (bulk), and  $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ .

Figure 6c shows the graph of Ollendorff's equation obtained from the theoretical value of Fe-6.5Si. In our work, we used  $N = \frac{1}{3} \exp(-3\eta)$  for the calculations, based on Anhalt et al. [47]. The permeability of the core with 80% packing fraction was 35 H/m as obtained by theoretical calculations, and 34 H/m as observed experimentally. The permeability of the core with 60% packing fraction was found to be 16 H/m through theoretical calculations, whereas

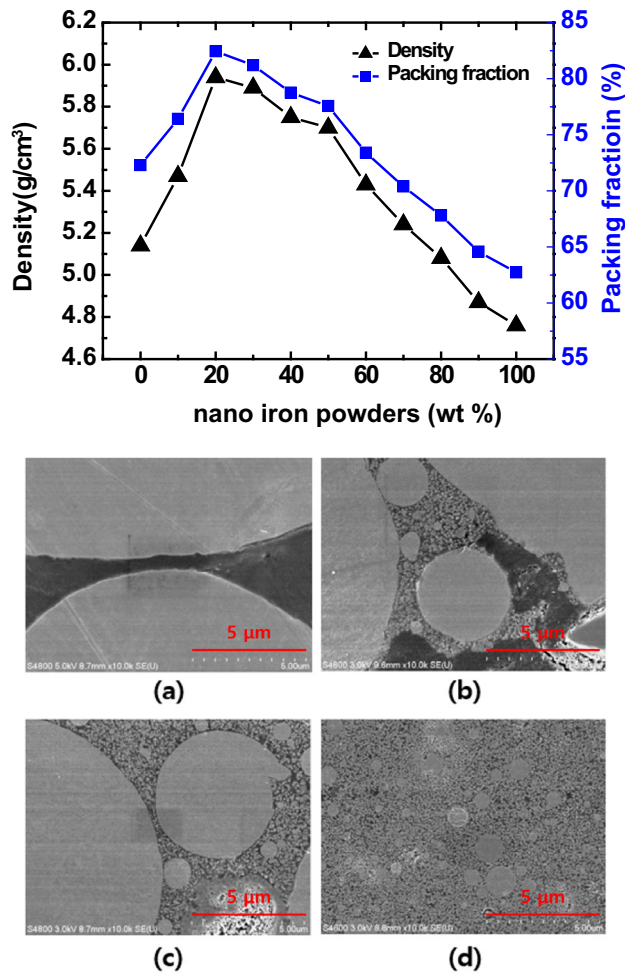
experimentally, a value of 12 H/m was obtained. Thus, the theoretical (obtained using Ollendorff's equation) and experimental values were almost identical. The measurement of the core loss while being used in a power loss application is one of the most important evaluation indices. Correspondingly, the core losses were measured at frequencies from 100 kHz to 3 MHz.

Figure 7 shows the frequency dependence of the core loss at a magnetic flux density of 10 mT. The core loss can be expressed as the sum of hysteresis loss, eddy-current loss, and residual loss, as shown in Eq. 2:

$$P_{tot} = P_h + P_e + P_r \cong P_h + P_e \cong W_h f + W_e f^2, \quad (2)$$

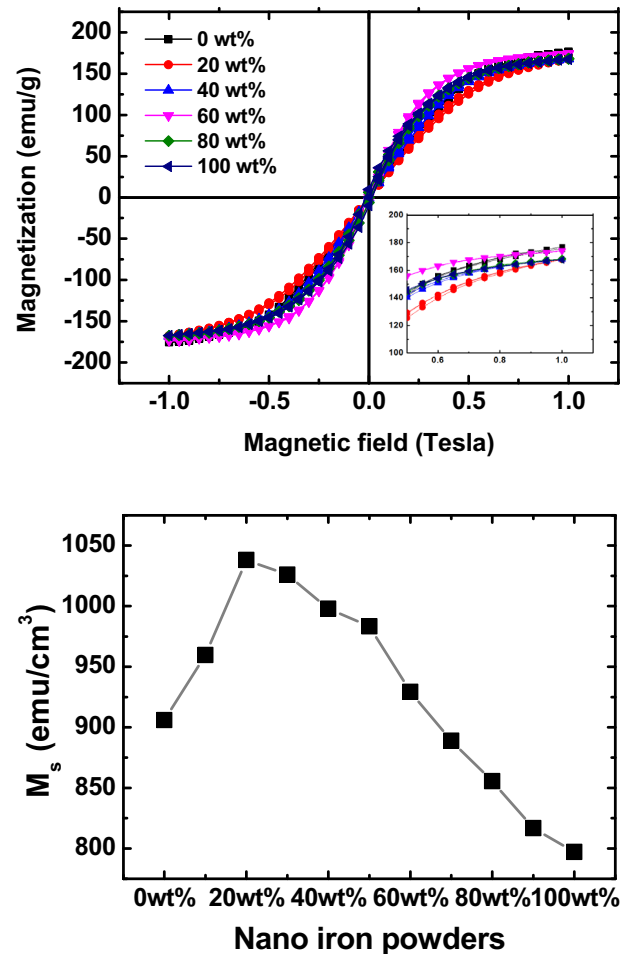
where  $P_h$  is the hysteresis loss,  $P_e$  is the eddy-current loss,  $P_r$  is the residual loss,  $W_h$  is the hysteresis loss constant, and  $W_e$  is the eddy-current loss constant. The residual loss does not affect the core and is negligible [15, 48, 49]. Therefore, only hysteresis loss and eddy-current loss are shown in Fig. 7.

As evident from Eq. 2 and Fig. 7, the hysteresis loss increases proportionally with the increasing



**Fig. 4** **a** Density and packing fraction, **b** Cross-section of Fe-6.5Si and nano iron powder toroidal core

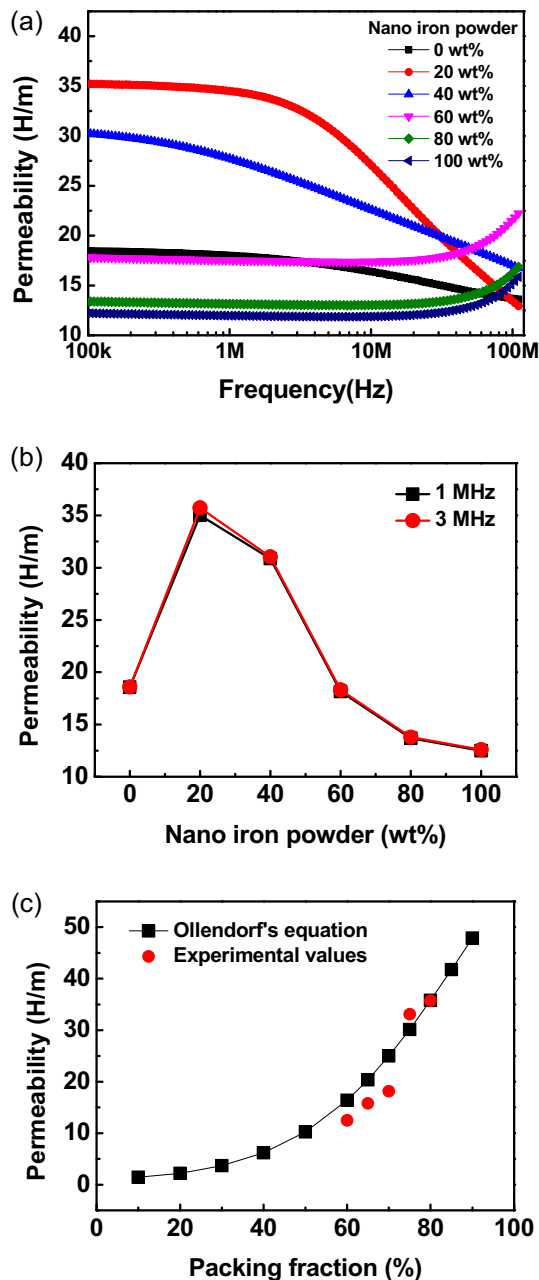
frequency. In our work, the hysteresis loss and eddy-current loss data were obtained via curve fitting. In the low-frequency range, the hysteresis loss and eddy-current loss are similarly low, or the eddy-current loss is lower. However, in the high frequency range, the eddy-current loss becomes the main factor that contributes to the total core loss. Therefore, reducing the eddy-current loss with the increasing driving frequency of electronic devices is critical. Even though the 0 wt% sample showed the lowest core loss, it exhibited an extremely low permeability compared to the other samples. Such a low permeability is not sufficient for power inductor applications. In contrast, for the other samples, the packing fraction was inversely proportional to the core loss. For the 20 wt% sample, which exhibited the highest packing fraction, showed a core loss of 200.35 mW/cm³ at 1 MHz.



**Fig. 5** **a** Hysteresis loop of core with nano iron powder, **b** magnetization value per unit volume at external magnetic field 1T

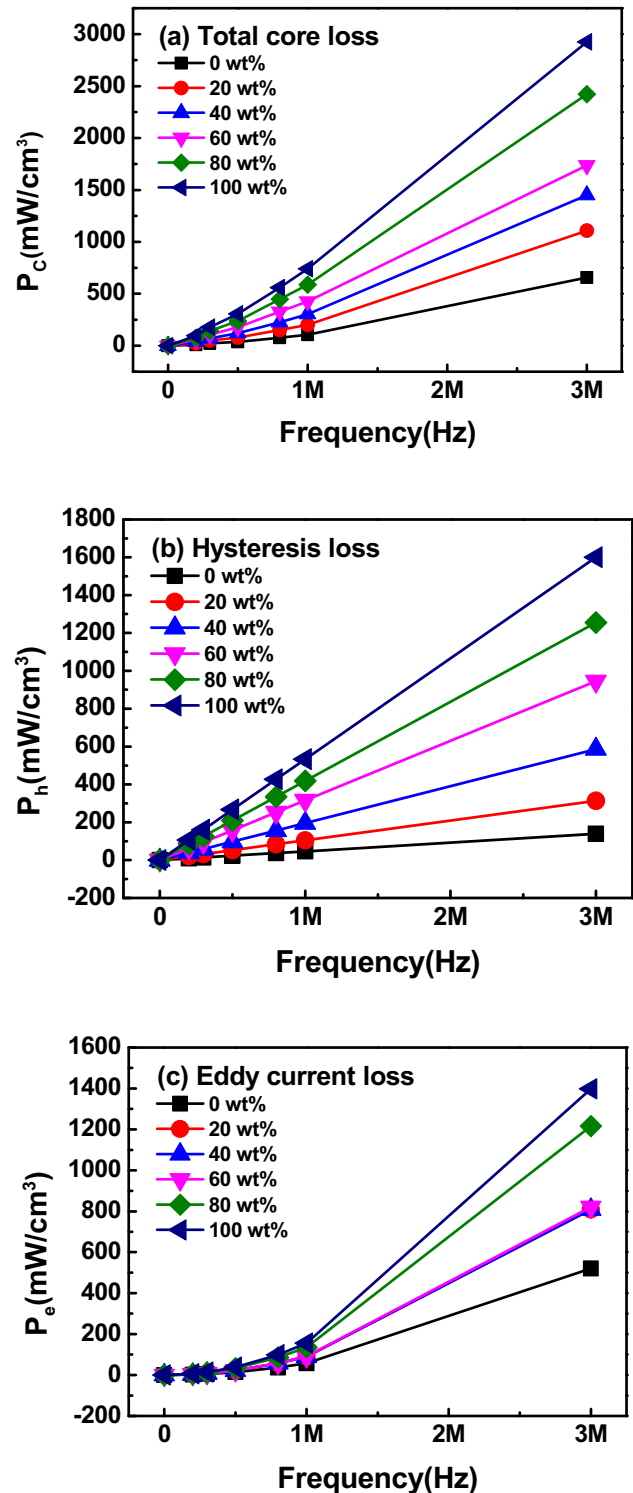
#### 4 Conclusion

In this study, we investigated feasible methods to improve the permeability of SMC cores. Cores with different ratios of Fe-6.5Si and nano Fe powders were prepared, and their AC and DC magnetic properties were investigated in the megahertz frequency range. It was found that the microstructure transformation, resulting from the improved core packing fraction, could improve the permeability as well as reduce the core loss at high frequencies. The maximum density and packing fraction were observed in the 20 wt% nano Fe powder core, which also showed the maximum permeability (35 H/m) and saturation magnetization. Using the Ollendorff's equation, we could estimate the permeability of the cores and prove that the permeability is proportional to the packing fraction of the core by comparing the experimental and



**Fig. 6** **a** Permeability of composites between 100 kHz and 110 MHz, **b** permeability of composite at 1 MHz and 3 MHz, **c** permeability of Ollendorf's equation and experimental data

calculated data. The 20 wt% sample with the best magnetic properties also showed a relatively low core loss of 200.35 mW/cm<sup>3</sup>. Through this study, we found that the core properties such as permeability and core loss can be improved through microstructural deformation as well as by improving the intrinsic properties of the soft magnetic materials used in the core.



**Fig. 7** **a** Total core loss, **b** hysteresis loss, **c** eddy-current loss of toroidal core as a function of operating magnetic flux at 10 mT, in 100 kHz to 3 MHz

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## Author contributions

MYL contributed toward methodology, preparation and writing of the original draft, and data curation. YJC contributed toward methodology, writing, reviewing, and editing of the manuscript. SHL contributed toward conceptualization, data analysis, preparation and writing of the original draft, data curation, and visualization. JHA contributed toward methodology, writing, reviewing, and editing of the manuscript. BWL contributed toward project administration, funding acquisition, and supervision.

## Data Availability

All data generated or analyzed during this study are included in this manuscript.

## Declarations

**Conflict of interest** The authors declare no competing interests.

**Ethical approval** We confirm that our manuscripts comply with ethical standards, and we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

**Research data policy** Our manuscript supports the Type 3 Research Data Policy.

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